Illeana Gomez Leal, PhD

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https://www.researchgate.net/profile/Illeana-Gomez-Leal-2

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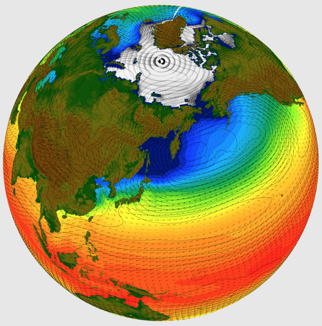
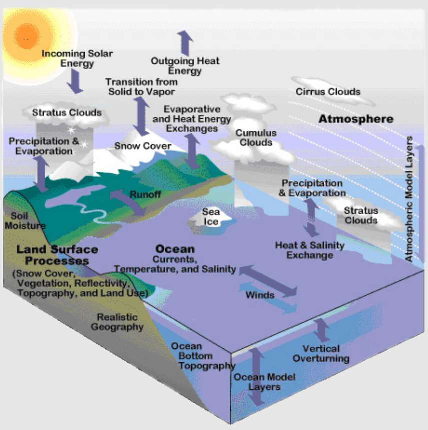
Research interests

- Global climate models
- Atmospheric dynamics
- Habitability
- Planetary and exoplanetary atmospheres
- Paleoclimate and climate change
- Satellite observations, planetary imaging, 3D mapping.





Research Experience

2015-2026 — **Research Associate** — Cornell University, Ithaca, USA
- https://orcid.org/0000-0003-2254-5936
- Cornell Center for Astrophysics and Planetary Science (CCAPS)/ Earth and Atmospheric Sciences (EAS)
- Global climate models (GCMs).
- Study of the habitability conditions of Earth-like planets and Super-Earths.
- Atmospheric dynamics of Earth-like planets and Super-Earths.

2014 — **Research Associate** — MPIA, Heidelberg, Germany
- Variation of the thermal light curves of Earth-like planets and Super-Earths using GCMs.
- Planet and Star Formation Department (PSF) at the Max-Planck Institut für Astronomie (MPIA): http://mpia.de/Public/menu_q2.php?../PSF/contact.php

2009-2013 — **PhD. in Astrophysics** — CNRS/Université de Bordeaux, France
- Spectrophotometry of the infrared emission of Earth-like planets. Advisor: Dr. Franck Selsis.
- Laboratoire d'Astrophysique de Bordeaux. Université de Bordeaux: https://tel.archives-ouvertes.fr/file/index/docid/962379/filename/GOMEZ_ILLEANA_2013_RED.pdf

Education

2013 — Ph.D. in Astrophysics — CNRS/Université de Bordeaux, France
- Thesis topic: *Spectrophotometry of the infrared emission of Earth-like planets*. Advisor: Dr. Franck Selsis. https://tel.archives-ouvertes.fr/tel-00962379/.

2009 — Postgraduate Certificate in Education — Univ. Alfonso X El Sabio, Spain
- https://www.uax.es/en/get-to-know-uax.html

2009 — M.Sc. in Astrophysics — Universidad de la Laguna, Spain
- Master thesis : *Photometric variability of the disk-integrated thermal emission of the Earth*. Advisor: Enric Pallé. https://www.ull.es/en/masters/astrophysics/

2007 — B.Sc. in Physics — Universidad Autónoma de Madrid, Spain
- Major in Theoretical Physics. https://www.uam.es/uam/grado

Awards

2009-2013 — International Graduate Fellowship — IAC, Spain/CNRS, France
- International Fellowship (PhD. Program) from the Instituto Astrofisico de Canarias (IAC), Spain and the Centre National de la Recherche Scientifique (CNRS), France.

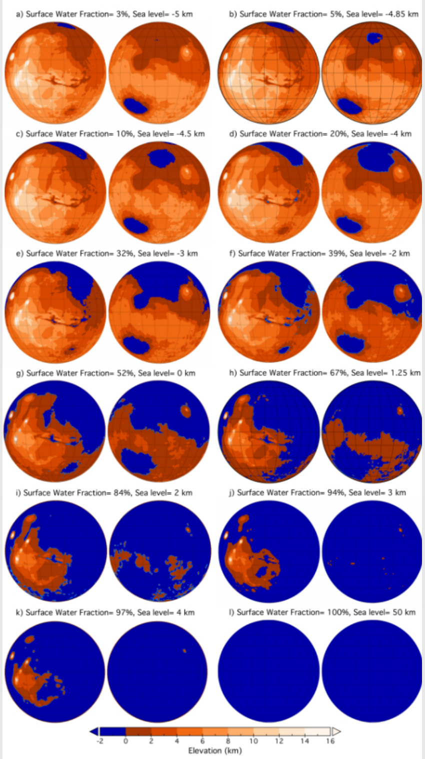
2007 — ESAC Trainee — European Space Astronomy Centre (ESAC), Spain
- Study of the reddening correction from the 2175 å interstellar dust absorption feature. Advisor: Dr. Carmen Morales-Duran.https://www.cosmos.esa.int/web/esac-trainees/trainees-2007

Skills

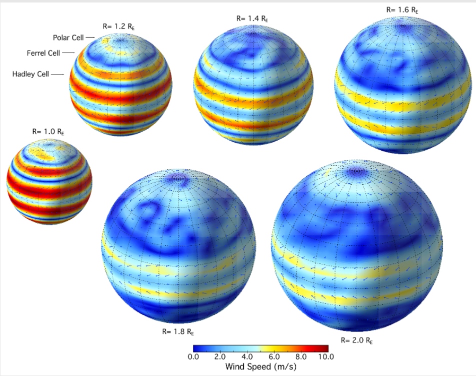
Programming languages
- Python, Fortran, IDL, Bash, Unix, LaTeX.

Global Climate Models
- LMDZ (Paris, France), PlaSim (Hamburg, Germany).

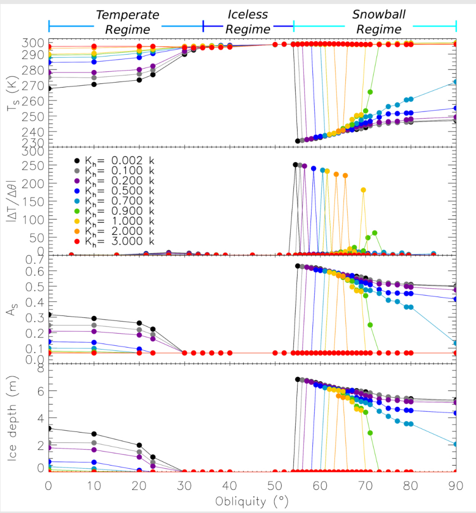
Languages
- Fluent in English, French, Italian, Spanish. Basic German.



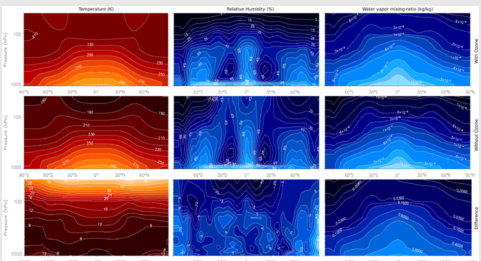
Habitability on Mars-size planets



Circulation of Super-Earth analogs



Climate sensibility to obliquity



Climate sensitivity to Ozone

Publications

2026 **Gomez-Leal, I.** *Water loss on Earth-like planets: evolution of irradiance and sea level.* In prep.

2026 **Gomez-Leal, I.**, Kaltenegger, L., Lucarini, V. & Lunkeit, F. *Climate sensitivity to obliquity and ocean heat transport on Earth-like planets.* In prep.

2026 **Gomez-Leal, I.** *Habitability of Mars-size planets: the importance of water content and surface albedo*. Submitted to Life, (ISSN 2075-1729; CODEN: LBSIB7). In review.

2026 **Gomez-Leal, I.**, Kaltenegger, L., & Lucarini V. *Cool Super Earths: a larger size reduces circulation and temperature* Submitted to ApJ. In review.

2020 Poggiali, V., Hayes, A. G., Mastrogiuseppe, M., Le Gall, A., Lalich, D., **Gomez-Leal, I.**, & Lunine, J. I. *The bathymetry of Moray sinus at Titan's Kraken Mare.* Journal of Geophysical Research: Planets, 125, e2020JE006558. `https://doi.org/10.1029/2020JE006558`

2019 **Gomez-Leal, I.**, Kaltenegger, L., Lucarini, V., & Lunkeit, F. 2019. *Climate sensitivity to ozone and its relevance on the habitability of Earth-like planets*. Icarus, 321, 608. `https://doi.org/10.1016/j.icarus.2018.11.019`

2018 **Gomez-Leal, I.**, Kaltenegger, L., Lucarini, V., & Lunkeit, F. 2018. *Climate Sensitivity to Carbon Dioxide and the Moist Greenhouse Threshold of Earth-like Planets under an Increasing Solar Forcing*. The Astrophysical Journal, 869, 2. `https://doi.org/10.3847/1538-4357/aaea5f`

2016 **Gomez-Leal, I.**, Codron, F., & Selsis, F. 2016. *Thermal light curves of Earth-like planets: 1. Varying surface and rotation on planets in a terrestrial orbit*. Icarus, 269, 98. `https://doi.org/10.1016/j.icarus.2015.12.050`

2015 Giovanna Tinetti, G., Drossart, P., Eccleston, P., and 352 coauthors, including **Gomez-Leal, I.** 2015. *The EChO science case*. Experimental Astronomy, 40, 329. `https://doi.org/10.1007/s10686-015-9484-8`

2012 **Gomez-Leal, I.**, Pallé, E., and Selsis, F. 2012. *Photometric Variability of the Disk-integrated Thermal Emission of the Earth*. The Astrophysical Journal, 752, 28. `https://doi.org/10.1088/0004-637X/752/1/28`

Selected contributions to Conferences

2020 ⋆ Talk.- P020-02-*The Importance of the Ocean Size on the Climate and the Water Loss of Mars-size planets*. AGU Fall Meeting, 1-17 December. `https://ui.adsabs.harvard.edu/abs/2020AGUFMP020...02G/abstract`

2017 ⋆ Talk.- Diversity of Planetary circulation regimes in our solar system and beyond. École de Physique, Les Houches, France.

Advisory Research and teaching experience

2015-2016 ⋆ Supervision of Research Experience for Undergraduates (REU) students and Graduate students at Cornell University.

2014 ⋆ Supervision of the Bachelor thesis of Nicolas Brod Bach and Anouk Ehreiser, students at Heidelberg University.

Science Vulgarization

2020-2021 Articles for the magazines: *L'Astronomie* (France), *Astronomia* (Spain), and *Cosmo* (Italy).